

Exploring the Different Types of Prompts

Straightforward • Tabular Format • Preceding Question • Missing Word

Course: Prompt Engineering & AI Applications | Unit 2 — Prompt Types

2 Scenarios per Prompt Type | With Examples, Use Cases & Output Notes

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AIM

To write prompts for four different prompt types — Straightforward, Tabular Format, Preceding Question, and Missing Word — covering 2 different use cases for each type, with reference to Unit 2 prompt structures and Unit 5 use cases.

Overview of Prompt Types

Prompt Type	Definition	Key Characteristic	Best Used For
Straightforward	Direct, clear instructions with no ambiguity	Single, focused request	Definitions, lists, quick facts
Tabular Format	Requests data or comparisons presented in table form	Structured comparative output	Comparisons, feature lists, pros/cons
Preceding Question	A context-setting question followed by the main task	Two-part: question + elaboration	Explanations, cause-effect, analysis
Missing Word	A sentence with a blank for the AI to fill in	Cloze-style completion	Recall, concept testing, fill-in tasks

1. Straightforward Prompts

Definition: A straightforward prompt is a clear, direct, single-request instruction that leaves no ambiguity. The AI is expected to respond precisely without extra reasoning steps. It is the simplest form of prompting and is ideal for factual or definitional queries.

Given Examples (from Experiment):

Example 1	"Define photosynthesis in one sentence."
Example 2	"List three advantages of electric vehicles."

Use Case Prompts:

UC 1	Prompt: "Define Artificial Intelligence in one sentence suitable for a school student." Expected Output / Note: <i>AI should return a single, simple sentence — e.g., "Artificial Intelligence is the ability of machines to perform tasks that normally require human intelligence, such as learning and problem-solving."</i>
UC 2	Prompt: "List five real-world applications of Machine Learning in healthcare." Expected Output / Note: <i>AI should return a numbered or bulleted list of 5 specific ML applications: e.g., disease diagnosis, drug discovery, patient risk prediction, medical imaging analysis, personalized treatment.</i>

2. Tabular Format Prompting

Definition: Tabular Format Prompting explicitly asks the AI to organize its response in a structured table with rows and columns. It is ideal for comparisons, feature listings, and side-by-side analysis of multiple items or concepts.

Given Examples (from Experiment):

Example 1	"Compare and contrast AC and DC current in a table."
Example 2	"Provide a table listing five programming languages, their paradigms, and one use case each."

Use Case Prompts:

UC 1

Prompt:

"Create a comparison table of Supervised Learning, Unsupervised Learning, and Reinforcement Learning with columns: Definition, Algorithm Examples, and Real-World Application."

Expected Output / Note:

AI should produce a 3-row, 4-column table covering each ML type with precise definitions, algorithms (e.g., SVM, K-Means, Q-Learning), and applications (e.g., spam detection, customer segmentation, game AI).

UC 2

Prompt:

"List the top 5 AI platforms (ChatGPT, Gemini, Claude, Copilot, Llama) in a table showing: Developer, Release Year, Strengths, and Best Use Case."

Expected Output / Note:

AI should return a 5-row table with accurate developer names, years, and concise strength/use-case descriptions for each platform.

Sample Output Table (for UC 2):

Platform	Developer	Year	Key Strength	Best Use Case
ChatGPT	OpenAI	2022	Versatile, conversational	General Q&A;, coding
Gemini	Google	2023	Multimodal, fast	Search integration
Claude	Anthropic	2023	Safety-focused, accurate	Edu. summarization
Copilot	Microsoft	2023	Office integration	Document drafting
Llama	Meta	2023	Open-source, customizable	Research, fine-tuning

3. Preceding Question Prompting

Definition: Preceding Question Prompting uses a context-setting question first, followed by the main instruction. The initial question primes the AI to think in a specific direction before elaborating. It is effective for analytical, explanatory, and cause-effect responses.

Given Examples (from Experiment):

Example 1	"Why is climate change a global concern? Explain how greenhouse gases contribute to global warming."
Example 2	"How do vaccines work? Describe the process of immunization in simple terms."

Use Case Prompts:

UC 1	Prompt: "What is a neural network? Explain how it learns from data using the concept of backpropagation, with an example from image recognition." Expected Output / Note: <i>The preceding question "What is a neural network?" sets context. The AI then elaborates on backpropagation and applies it to image classification — a two-part, logically connected response.</i>
UC 2	Prompt: "Why is data preprocessing important in machine learning? Describe the steps involved in cleaning and preparing a raw dataset before training a model." Expected Output / Note: <i>Opening question establishes importance; follow-up drives a step-by-step explanation covering missing value handling, normalization, encoding, and train-test split.</i>

Structure of Preceding Question Prompt:

PART 1: Context Question	PART 2: Main Instruction
Sets the topic/context for the AI e.g. "What is a neural network?"	Asks for elaboration or explanation e.g. "Explain how backpropagation works..."

4. Missing Word Prompting

Definition: Missing Word Prompting (also called Cloze Prompting) presents a sentence with one or more blanks (____) that the AI must fill in. It tests knowledge recall, contextual understanding, and concept completion. Widely used in educational assessments and knowledge testing.

Given Examples (from Experiment):

Example 1	"The capital of France is ____."
Example 2	"In photosynthesis, plants absorb sunlight to produce ____."

Use Case Prompts:

UC 1	Prompt: "In machine learning, the process of adjusting model weights based on the error between predicted and actual outputs is called ____." Expected Output / Note: <i>Expected Answer: Backpropagation. Tests understanding of neural network training. The blank targets a specific technical term, making it effective for concept recall exercises.</i>
UC 2	Prompt: "A blockchain is a ____ ledger that records transactions across a network of computers, where each ____ contains data, a timestamp, and a cryptographic ____." Expected Output / Note: <i>Expected Answers: decentralized / block / hash. Multiple blanks test deeper comprehension of the blockchain concept, covering three key terms in context.</i>

Missing Word Answer Key Table:

#	Incomplete Sentence	Missing Word(s)	Concept Tested
1	The capital of France is ____.	Paris	Geography recall
2	Plants absorb sunlight to produce ____.	Glucose / Food	Photosynthesis
3	Adjusting model weights based on error is called ____.	Backpropagation	ML Training
4	Blockchain is a ____ ledger with ____ and ____.	decentralized / block / hash	Blockchain basics

Summary Comparison Table — All 4 Prompt Types

Prompt Type	Structure	Use Case 1 Topic	Use Case 2 Topic	Output Style
Straightforward	Direct single request	Define AI for school students	List ML applications in healthcare	Sentence / Bullet list
Tabular Format	Request data in table form	Compare ML types (table)	AI platforms comparison (table)	Structured table
Preceding Question	Question + elaboration request	What is neural network? (backprop)	Why data preprocessing? (steps)	Two-part explanation
Missing Word	Sentence with blank(s)	Backpropagation cloze fill	Blockchain multi-blank fill	Keyword completion

Result

The experiment successfully demonstrated the four major prompt types — **Straightforward, Tabular Format, Preceding Question, and Missing Word** — with 2 distinct use cases each. Each technique was shown to serve a specific purpose: Straightforward prompts for quick facts, Tabular prompts for organized comparisons, Preceding Question prompts for in-depth explanation, and Missing Word prompts for concept recall and testing. Understanding these prompt types equips students with the foundational skills to design effective AI interactions across diverse domains.